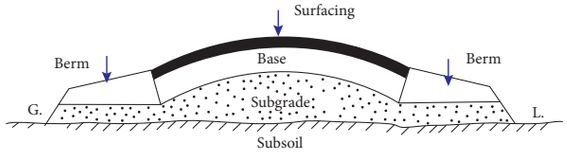
alignment, cross section elements, sight distances and intersections.

The geometrics of highway should be designed to provide optimum efficiency in traffic operations with maximum safety at reasonable cost.

6.3.1 Road Structure



Road Structure

1. **Subgrade:** It is the natural foundation on which the entire road structure rests. The life of road depends primarily on stable and dry subgrade.
2. **Formation:** It is the top surface of the subgrade.
3. **Sub-base:** It is the course laid between the base course and subgrade. It is used only when the bearing capacity of subgrade is poor or when the subgrade has poor drainage properties.
4. **Base course:** It is the structural foundation of the road. It is also termed as soling. The function of a road base is to transmit load of traffic from the surfacing to the subgrade. It should possess structural stability and should be of sufficient thickness to develop a good bond with the surfacing.
5. **Wearing course:** The topmost layer on which the traffic directly travels is known as wearing course. The road surfacing will provide a smooth and stable running surface. The surfacing should be impervious and should protect the base and the subgrade from the action of weather and rain water.

6.3.2 Width of Road

The width of pavement or carriage way depends on the width of traffic lane and number of lanes. The carriage way intended for one line of traffic movement may be called as a **traffic lane**. The lane width is determined on the basis of the width of vehicle and the minimum side clearance provided for the safety. When the side clearance is increased there is an increase in speed of the vehicles and hence increase in the capacity of the pavement. A width of 3.75 m is considered desirable for a road having single lane for vehicles of maximum width 2.44 m. For pavement having two or more lanes, width of 3.5 m per lane is sufficient.

6.3.3 Right of Way

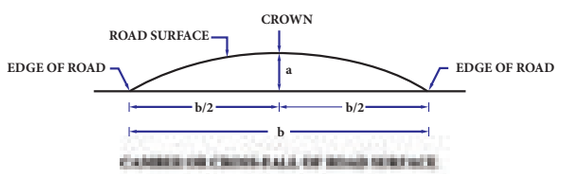
Right of way is the area of land acquired and reserved along its alignment for construction and development of a highway.

6.3.4 Highway Cross Section Elements

i. Camber Or Cross Slope Or Cross Fall

The cross-section of a road surface shows the convexity upwards and the highest point on the curved surface is known as crown.

Camber is defined as the slope of the line joining the crown and the edge of the road surface.



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